

Yidi Zhang

zhangyidi.lily@gmail.com | zhangyidi.tech | github.com/sorceressyidi

Education

Zhejiang University, Chu Kochen Honors College
Bachelor of Science, Computer Science and Technology
Advisor: Prof. Sida Peng

Sep. 2022 – June 2026
(Expected)

- GPA: 4.12/4.30 (91.27/100), Ranking: Top 10%
- **Selected Coursework:** Mathematical Analysis I(H) (100/100), Linear Algebra II(H) (99/100), Introduction to Natural Language Processing(97/100), Advanced Data Structures and Algorithm Analysis(94/100)

University of California, Berkeley
Exchange Student

Jan. 2025 – May 2025

- **Selected Coursework:** COMPSCI 280: Computer Vision (Graduate Level) (A)

Publications

[1] Yash Jangir, **Yidi Zhang**, Kashu Yamazaki, Chenyu Zhang, Kuan-Hsun Tu, Tsung-Wei Ke, Lei Ke, Yonatan Bisk, and Katerina Fragkiadaki. **ROBOTARENA ∞ : Unlimited Robot Benchmarking via Real-to-Sim Translation**. In International Conference on Learning Representations (ICLR), 2026. [Under Review] [Project Page] [arXiv]

Experience

Carnegie Mellon University
Research Intern, Advisor: Prof. Katerina Fragkiadaki

Jan. 2025 – Present

- **ROBOTARENA ∞ : Unlimited Robot Benchmarking via Real-to-Sim Translation**—Developed a scalable benchmark for real-world-trained robot policies, converting video demonstrations into simulated environments with automated vision-language scoring and human preference feedback, enabling reproducible, robust, and large-scale evaluation of robot generalist performance
- Contributed to full pipeline development, including scalable real-to-sim transformation automation and human preference feedback design, with supporting web infrastructure for large-scale benchmarking.

Zhejiang University
Research Intern, Advisors: Prof. Sida Peng & Prof. Xiaowei Zhou

Aug. 2023 – Jan. 2025

- Gained comprehensive knowledge in 3D computer vision through advanced coursework including ML 4360, CS231n, and EECS 498-007.
- **4D Gaussian Splatting Research**—Improved the optimization framework with depth-aware supervision for better spatio-temporal consistency and geometric fidelity, and enabled efficient streamable rendering with significantly reduced memory usage.

Projects

DREAMoR: Diffusion-based Reconstruction and Motion Prior

Project Page

- Final project for UC Berkeley COMPSCI 280: Computer Vision (Graduate Level). Team members: Sihan Ren, Jiashen Du, Jingfeng Yang, **Yidi Zhang**.
- Proposed **DREAMoR**, a diffusion-based motion prior framework for reconstructing physically plausible human motion from monocular sequences degraded by occlusion or noise.
- Developed a latent-space diffusion prior with multi-step DDIM denoising and score distillation optimization, integrating temporal consistency and physics-based constraints to enhance realism and robustness.

Chain-of-Thought Reasoning with Large Language Models

GitHub

- Developed and implemented innovative techniques including Multi-vote, LoRA fine-tuning, and Multi-path step-aware inference to enhance reasoning and problem-solving capabilities of large language models.
- Designed comprehensive experimental framework to evaluate and compare different chain-of-thought

prompting strategies.

miniOS: Operating System Development

- Developed a minimal operating system implementing key functionalities including custom bootloader and kernel initialization, memory management, process scheduling and context switching, filesystem design, and device driver integration using C++ and RISC-V assembly.

Skills

Programming: Python, C/C++, RISC-V Assembly, Bash, Verilog, HTML

Tools: Git, Colmap, MeshLab, LaTeX

Languages: Chinese (Native), English (TOFEL:108 GRE 331)

Honors & Awards

First-Prize Scholarship of Zhejiang University Fall 2024

- Awarded by Zhejiang University, Top 2%

Zhejiang Provincial Government Scholarship Fall 2023

- Awarded by Ministry of Education of Zhejiang Province, Top 2%

Scholarship for Exceptional Students in Fundamental Subjects Fall 2023

- Awarded by Zhejiang University, Top 10% in department